

Exponent-Sequence and Newton-Basis Frontiers for the Fabius–Rvachev System

A universal spectral–cumulant dictionary, certified q -Richardson tomography, positive-definite Pascal quotients, automatic lobe signs, and Gaussian boundary layers

Frontier research report based on the complete recursive `*.tex` corpus under
Analysis/FabiusFunction/docs

Pinned repository snapshot: 27 August 2026

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Abstract

The audited PROVEIT corpus already contains the classical Fabius and Rvachev functions, exact dyadic arithmetic, the full Thue–Morse formula atlas, Gaussian q -binomial interpolation, all-orders endpoint expansions, lower-branch Lambert- W coordinates, Bell- and Bernoulli-polynomial coefficient calculus, sharp Fourier decay, the integer-zero spectrum of the Rvachev transform, a spectral determinant, rational arithmetic rays, half-integer quadrature identities, logarithmic q -Richardson acceleration, and a Pascal-weighted hierarchy of compactly supported densities. This report avoids repackaging those results. Instead it identifies a single master object that contains them as special coordinates and exposes several new deductions.

For a nonnegative integer exponent sequence $a = (a_h)_{h \geq 0}$ with $A(1/2) < \infty$, where $A(q) = \sum_{h \geq 0} a_h q^h$, define

$$\Phi_a(z) = \prod_{h \geq 0} \operatorname{sinc}\left(\frac{z}{2^h}\right)^{a_h}, \quad \operatorname{sinc} z = \frac{\sin \pi z}{\pi z}.$$

It is the characteristic function of an explicit compactly supported convolution of uniform variables. We prove a universal dictionary in which the same generating function A controls: the integer-zero multiplicity $m_a(n) = \sum_{h \leq \nu_2(n)} a_h$; the spectral zeta $\mathcal{Z}_a(s) = \zeta(s)A(2^{-s})$; every even cumulant $\kappa_{2j} = B_{2j}A(4^{-j})/(2j)$; the Fourier-lobe sign; exact digital zero counts; and the order of Prouhet cancellation of each finite sign block. The zero divisor, the spectral zeta quotient, and the complete even-cumulant sequence each determine a uniquely.

The cumulant samples $A(4^{-m})$ admit a new exact reconstruction mechanism. Lagrange extrapolation at geometric nodes has Gaussian- q -binomial weights, an exact transformed series, fixed alternating sign, bounded ℓ^1 condition number, and certified integer brackets. Thus Richardson extrapolation becomes a stable tomography algorithm for the hidden dyadic exponent sequence.

For polynomial exponent sequences, the Newton expansion $P(h) = \sum c_r \binom{h}{r}$ gives the finite factorization $\Phi_P = \prod_r \Phi_{r+1}^{c_r}$ through the audited Pascal–Rvachev hierarchy. Negative Newton coefficients are allowed: whenever $P(h) \geq 0$ on the nonnegative integers, the resulting alternating quotient is nevertheless entire and positive definite. The even family

$$P_d(h) = \binom{h-1}{d}, \quad d = 0, 2, 4, \dots,$$

is especially striking. It satisfies

$$\Psi_d(z) = \operatorname{sinc}(z)\Phi_{d+1}(z/2) = \prod_{r=0}^d \Phi_{r+1}(z)^{(-1)^{d-r}},$$

has support $[-1, 1]$, zero multiplicity $1 + \binom{\nu_2(n)}{d+1}$, and converges in total variation to the uniform law. Its transition across the two discontinuities of the limiting box density has a Gaussian boundary layer with an explicit Berry–Esseen rate. For $d = 2$ this supplies the concrete positive-definite quotient $\Phi_1 \Phi_3 / \Phi_2$.

Finally, a Mellin-reflection identity relates the endpoint Laplace transform directly to the reflected spectral zeta. For polynomial weights it explains the complete hierarchy of logarithmic powers and periodic coefficients, gives a generalized lower-Lambert saddle, and yields the leading inverse-endpoint law. Reasonable conjectures and a detailed research and Lean-formalization program conclude the report. Numerical experiments were independently implemented at high precision; the commented Python source and generated data accompany the paper.

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1 Audit boundary, source stratification, and novelty discipline

1.1 Pinned corpus

The repository changed several times on the audit date, so the source boundary is the exact commit printed on the title page. Its recursive documentation tree contains 67 distinct `.tex` paths: five living synthesis/reference documents, thirty-two frozen historical sources, twenty-three active frontier drafts (including twelve Fourier-decay dossiers), and seven transcribed or corrected source papers. Every path is listed in [section A](#). Historical revisions were read as mathematical lineages rather than as independent evidence; the archive’s provenance ledger identifies what was superseded, merged, or deliberately frozen.

The living primary exposition separates the bounded distribution function, the compactly supported Rvachev bump, and the signed global extension, and confines itself to claims with matching Lean declarations. The consolidated frontier volume and its current descendants contain the non-formalized but mathematically developed material. In particular, the audit treated the following as prior art rather than as discoveries of this report:

- the integer zero set and multiplicity $1 + \nu_2(n)$ of the classical Rvachev transform;
- its spectral determinant, spectral zeta, heat trace, local zero jets, and symmetric q -integer deformation;
- the full signed Thue–Morse formula atlas, Prouhet cancellation, Mahler products, rarefied sums, Mellin–Dirichlet theory, and automatic-sequence machinery;
- exact dyadic values of F , half-base q -Pochhammer and Gaussian-binomial formulas, Lagrange interpolation, and several existing Richardson transforms;
- the corrected lower-Lambert endpoint expansion, its nonconstant periodic factor, all-order Bell-polynomial coefficient construction, and the inverse Fabius expansion;
- sharp geometric, arithmetic, metric, and envelope results for the infinite sinc product;
- cyclotomic rational rays, half-integer spectral identities, exact zero jets, and the Pascal hierarchy $\Phi_r(z) = \prod_h \text{sinc}(z/2^h)^{\binom{h}{r-1}}$;
- logarithmic q -Richardson acceleration of truncated sinc products and phase-locked Richardson extraction of the endpoint periodic term;
- positive Gaussian and Lobatto tail quadrature, exact ordinary-prefix error laws, generalized gamma-convolution duality, and the existing Gaussian rank limit for the Pascal hierarchy.

1.2 What is new relative to the pinned tree

Targeted path, phrase, and formula searches found no counterpart of the following theorem package in the audited snapshot:

1. the arbitrary nonnegative exponent-sequence monoid and its single generating-function dictionary for zero spectra, cumulants, digital signs, and endpoint Mellin transforms;
2. exact digital zero-count defects and generalized Prouhet order for an arbitrary exponent parity word, including the first surviving power moment;
3. equivalence and constructive recovery of the exponent sequence from either the zero divisor or all even cumulants;
4. certified Gaussian- q -binomial Richardson tomography of the exponent coefficients;
5. the finite-refinement-rank classification by rational ordinary generating functions;

6. the Newton-basis factorization of every nonnegative integer-valued polynomial weight, including positive-definite alternating quotients of Pascal transforms;
7. the normalized family $P_d(h) = \binom{h-1}{d}$ for even d , its sparse zero spectrum, automatic sign selector, exact total-variation law, and Gaussian boundary layer;
8. the general spectral-endpoint Mellin reflection formula $\mathcal{M}[\log L_a](w) = -\Gamma(w)\zeta(w+1)\mathcal{Z}_a(-w)/\zeta(-w)$;
9. the polynomial-weight universality theorem and its generalized Lambert- W_{-1} endpoint coordinate.

A targeted external search located the classical ingredients separately—the original Fabius–Rvachev analysis and arithmetic, lacunary Thue–Morse products, automatic sequences, Berry–Esseen bounds, rational and integer-valued generating functions, Mellin harmonic sums, exponential Tauberian transfer, and Lambert inversion [4, 5, 1, 12, 2, 10, 11, 8, 7, 6]—but not this combined classification. Accordingly, “new” below always means new to the pinned corpus and apparently nonstandard; it is not an assertion of exhaustive worldwide priority.

Proof-status convention. Results explicitly labeled **proved here** are proved in this manuscript. Results labeled **derived from audited corpus theorems** use a theorem already developed in the audited frontier corpus (most often the sharp Pascal-hierarchy Fourier or endpoint asymptotic) and then add a new deduction. Items labeled **conjectural/open** are conjectures or proposed programs.

A fourth, orthogonal label is now carried alongside these three. Wherever a statement of this report—or an exactly reusable core of one—already has a machine-checked counterpart in the repository’s Lean library, that declaration is cited inline in typewriter type—for instance `Fabius.weightedScaleMultiplicity`—together with its module under `Analysis/FabiusFunction/Lean/FabiusFunction/`. Roughly a dozen such citations appear below. In most of them the formal statement is *more general* than the one printed here (arbitrary base instead of 2, arbitrary coefficient ring or module instead of $\mathbb{R}$, arbitrary geometric ratio instead of $q \in (0, 1)$). Each citation says which. In particular the finite rational geometric-Lagrange kernel used in the Gaussian- q rows—closed weights, all residual moments and their sign, and the finite exact ℓ^1 norm—is formalized at arbitrary ratio $0 < q < 1$, so the $q = 1/4$ instance is covered. These labels concern representation only, not correctness: no result of this report is formalized end to end, and every result below that carries no such citation is unformalized. In particular the transformed infinite series, Richardson tomography and its analytic remainder, the infinite condition-number limit, the entire probabilistic layer, the entire analytic layer—Fourier envelopes for the Pascal hierarchy of rank at least two, Mellin contour displacement, exponential Tauberian transfer, Berry–Esseen for non-identically-distributed summands, total-variation and Wasserstein distances—and the automatic-sequence machinery have no Lean counterpart at all.

2 The exponent-sequence convolution monoid

2.1 Admissible sequences

Throughout, the normalized entire sinc is

$$\operatorname{sinc} z = \frac{\sin \pi z}{\pi z}, \quad \operatorname{sinc} 0 = 1. \quad (2.1)$$

It is the Fourier transform, under the kernel $e^{-2\pi izx}$, of the uniform probability density on $[-1/2, 1/2]$.

Definition 2.1 (Admissible exponent sequence). A sequence $a = (a_h)_{h \geq 0}$ is *admissible* if

$$a_h \in \mathbb{N}_0, \quad A(q) := \sum_{h \geq 0} a_h q^h, \quad A(1/2) < \infty.$$

Its generalized Rvachev transform is

$$\Phi_a(z) = \prod_{h=0}^{\infty} \operatorname{sinc}\left(\frac{z}{2^h}\right)^{a_h}. \quad (2.2)$$

For each h and $1 \leq j \leq a_h$, let $U_{h,j}$ be independent and uniform on $[-1/2, 1/2]$, and put

$$X_a = \sum_{h \geq 0} \sum_{j=1}^{a_h} 2^{-h} U_{h,j}, \quad R_a = \frac{1}{2} A(1/2). \quad (2.3)$$

Absolute convergence is deterministic because the sum of the radii is R_a .

Theorem 2.2 (Probability, analyticity, support, and convolution). *For every admissible a :*

1. X_a is supported on $[-R_a, R_a]$ and has characteristic function Φ_a ;
2. the product in (2.2) converges locally uniformly on $\mathbb{C}$, hence is entire;
3. $\Phi_{a+b} = \Phi_a \Phi_b$, and the corresponding probability law is the convolution of the laws for a and b ;
4. if $\sum_h a_h = \infty$, then X_a has a compactly supported C^∞ density; if only finitely many a_h are nonzero, its law is a finite uniform box spline.

All four assertions are **proved here**. Formally, only the classical case $a_h \equiv 1$ is represented: `Fabius .rvachevFourier_eq_product` (`FourierProduct.lean`) identifies the Fourier transform of the Rvachev up-function with the sinc product (2.2) at $a_h \equiv 1$, in the same normalization (2.1), and `Fabius .ProbabilityRepresentation.rvachevUp_eq_weightedSum_probability_global` (`ProbabilityRepresentation.lean`) is the matching dyadic-uniform-series representation, stated in the Fabius normalization $\sum_{h \geq 0} 2^{-h-1} V_h$ on $[0, 1]$, which is $(X_a + R_a)/2$ in the notation used here. Nothing about a general admissible a is formalized.

Proof. Independence gives the finite-prefix characteristic products. The variables are bounded and $\sum_{h,j} |2^{-h} U_{h,j}| \leq R_a$, so the finite sums converge almost surely and in every L^p , and their characteristic functions converge to that of X_a .

Locally, $\log \operatorname{sinc} w = \mathcal{O}(w^2)$. Since $\sum_h a_h 4^{-h} = A(1/4) < \infty$, the logarithms of the factors converge normally on compact sets avoiding a harmless choice of local branch; standard infinite-product theory then gives an entire limit. Addition of exponent sequences is multiplication of products, which is exactly convolution of independent sums.

If infinitely many individual factors occur, fix M of them. On the real axis, $|\operatorname{sinc}(\xi/2^h)| \leq \min(1, 2^h/(\pi|\xi|))$. The chosen M factors give $|\Phi_a(\xi)| \leq C_M(1 + |\xi|)^{-M}$, while all remaining factors have modulus at most one. Thus Φ_a is rapidly decreasing; Fourier inversion gives a C^∞ density. With finitely many factors one obtains a finite convolution of interval densities, hence a box spline. $\square$

Remark 2.3. The classical Rvachev transform corresponds to $a_h \equiv 1$. The audited Pascal hierarchy corresponds to $a_h = \binom{h}{r-1}$. The point of [theorem 2.2](#) is not merely that both products exist, but that the collection of all such laws is a commutative convolution monoid whose algebra is the coefficientwise additive monoid of exponent sequences.

3 Integer zeros, spectral zeta, digital defects, and lobe signs

3.1 The universal zero-divisor formula

Euler’s product for the normalized sinc gives

$$\operatorname{sinc} z = \prod_{k \geq 1} \left(1 - \frac{z^2}{k^2}\right).$$

An integer $n \neq 0$ is a zero of the scale- h factor precisely when $2^h \mid n$. This immediately yields the following master theorem.

Theorem 3.1 (Zero divisor and spectral zeta). *Let a be admissible and define*

$$m_a(n) = \sum_{h=0}^{\nu_2(n)} a_h, \quad n \geq 1. \quad (3.1)$$

Then

$$\Phi_a(z) = \prod_{n \geq 1} \left(1 - \frac{z^2}{n^2}\right)^{m_a(n)} \quad (3.2)$$

with locally uniform convergence. In particular, $\operatorname{ord}_{z=\pm n} \Phi_a = m_a(n)$. For $\Re s > 1$,

$$\mathcal{Z}_a(s) := \sum_{n \geq 1} \frac{m_a(n)}{n^s} = \zeta(s) A(2^{-s}). \quad (3.3)$$

Both statements are **proved here**. The multiplicity function (3.1) is formalized as `Fabiús.weightedScaleMultiplicity` (`WeightedScaleMultiplicity.lean`), and in two directions more generally than it is used here: the base 2 is replaced by an arbitrary b , and the weights a_h may lie in an arbitrary additive commutative monoid rather than in $\mathbb{N}_0$. The canonical product (3.2) is formalized only at $a_h \equiv 1$, as `Fabiús.rvachevFourierProduct_eq_canonical` (`SincCanonicalProduct.lean`), whose exponent is the classical $\nu_2(n) + 1$; the general- a product and the spectral zeta (3.3) are not formalized.

Proof. The scale- h sinc contributes one zero at n for every representation $n = 2^h k$ with $k \geq 1$, and it occurs with multiplicity a_h . Summing over $h \leq \nu_2(n)$ gives (3.1). Moreover

$$\sum_{n \geq 1} \frac{m_a(n)}{n^2} = \sum_{h \geq 0} a_h \sum_{k \geq 1} \frac{1}{(2^h k)^2} = \zeta(2) A(1/4) < \infty,$$

so the genus-zero canonical product converges normally. The same rearrangement for $\Re s > 1$ gives

$$\sum_{n \geq 1} \frac{1}{n^s} \sum_{2^h \mid n} a_h = \sum_{h \geq 0} a_h 2^{-hs} \zeta(s).$$

□

Spectral–probabilistic dual use of A . The spectral side samples A at 2^{-s} , while the probability side will sample it at 4^{-j} . The entire zero divisor and every even cumulant are therefore two transforms of one coefficient sequence, not merely analogous formulas.

3.2 Exact counting and a generalized digital defect

Let

$$M_a(N) = \sum_{n=1}^N m_a(n). \quad (3.4)$$

For $N = \sum_{j \geq 0} b_j(N)2^j$, define the tail weights

$$\tau_j(a) = \sum_{k \geq 1} a_{j+k} 2^{-k}. \quad (3.5)$$

Theorem 3.2 (Exact zero count and digit defect). *For every positive integer N ,*

$$M_a(N) = \sum_{h \geq 0} a_h \left\lfloor \frac{N}{2^h} \right\rfloor, \quad (3.6)$$

$$= NA(1/2) - \rho_a(N), \quad \rho_a(N) = \sum_{h \geq 0} a_h \left\{ \frac{N}{2^h} \right\}, \quad (3.7)$$

$$= NA(1/2) - \sum_{j \geq 0} b_j(N) \tau_j(a). \quad (3.8)$$

This is **proved here**. The floor form (3.6) is formalized as `Fabius.sum_range_weightedScaleMultiplicity` (`WeightedScaleMultiplicity.lean`), over the finite range $h < N$ that already contains every nonzero term, and again for an arbitrary base $b \neq 1$ and weights in an arbitrary additive commutative monoid, so the Lean statement is strictly more general than (3.6). The two defect forms (3.7)–(3.8) are not formalized.

Proof. Count the multiples of 2^h among $1, \dots, N$ and sum their contributions a_h . Subtracting the fractional parts from $N \sum_h a_h 2^{-h}$ gives (3.7). Finally,

$$\left\{ \frac{N}{2^h} \right\} = \sum_{j=0}^{h-1} b_j(N) 2^{j-h};$$

absolute convergence permits exchanging the sums and yields (3.8). $\square$

For $a_h \equiv 1$, one has $A(1/2) = 2$ and $\tau_j = 1$, recovering the audited identity $M_a(N) = 2N - s_2(N)$. For general a , the ordinary binary weight is replaced by a weighted suffix transform of the exponent sequence.

3.3 Lobe signs and generalized Thue–Morse characters

For $x \in (N, N+1)$, every canonical-product factor with index $n \leq N$ is negative and all remaining factors are positive. Hence

$$\text{sgn } \Phi_a(x) = (-1)^{M_a(N)}. \quad (3.9)$$

Modulo two, $\lfloor N/2^h \rfloor \equiv b_h(N)$, so the sign is a digit character.

Theorem 3.3 (Generalized lobe sign). *Put $\bar{a}_h = a_h \bmod 2$ and*

$$\varepsilon_a(N) = (-1)^{\sum_{h \geq 0} \bar{a}_h b_h(N)}. \quad (3.10)$$

Then, away from the integer zeros,

$$\boxed{\text{sgn } \Phi_a(x) = \varepsilon_a(\lfloor x \rfloor)}. \quad (3.11)$$

Moreover ε_a is a 2-automatic sequence if and only if the parity word $(\bar{a}_h)_{h \geq 0}$ is eventually periodic. Both statements are **proved here**. At $a_h \equiv 1$ the sign law (3.11) is formalized as `Fabiús.rvachevFourierProduct_eq_thueMorse_sign_mul_norm (ThueMorseLobeSign.lean)`, in the equivalent form $\Phi(x) = (-1)^{s_2(N)} \|\Phi(x)\|$ on the lobe $(N, N+1)$. The general parity character (3.10) is not formalized, and neither is the automaticity criterion, for want of any automatic-sequence or 2-kernel theory in the library.

Proof. The sign formula follows from (3.9), (3.6), and reduction modulo two. For the automaticity criterion, let S denote the left shift of the parity word. If $0 \leq r < 2^k$, binary concatenation gives

$$\varepsilon_a(2^k n + r) = \varepsilon_a(r) \varepsilon_{S^k a}(n). \quad (3.12)$$

Thus an eventually periodic parity word has only finitely many shifts and therefore gives a finite 2-kernel. Conversely, equality of two shifted sign sequences forces equality of the corresponding shifted parity words, because $\varepsilon_{S^k a}(2^j) = (-1)^{\bar{a}_{k+j}}$. A finite 2-kernel therefore implies a finite shift orbit of the parity word, which is equivalent to eventual periodicity. This is the present Walsh-character specialization of the standard finite-kernel characterization of automatic sequences [2]. $\square$

3.4 Finite sign products and exact Prouhet order

For $L \geq 0$, let

$$O_L(a) = \{0 \leq h < L : a_h \text{ is odd}\}, \quad \omega_L(a) = |O_L(a)|. \quad (3.13)$$

Theorem 3.4 (Generalized Prouhet block). *The length- 2^L sign polynomial factors as*

$$T_{a,L}(z) := \sum_{n=0}^{2^L-1} \varepsilon_a(n) z^n = \prod_{h=0}^{L-1} \left(1 + (-1)^{a_h} z^{2^h}\right). \quad (3.14)$$

Its zero at $z = 1$ has exact order $\omega_L(a)$. Consequently

$$\sum_{n < 2^L} \varepsilon_a(n) n^k = 0, \quad 0 \leq k < \omega_L(a), \quad (3.15)$$

and the first surviving power moment is

$$\sum_{n < 2^L} \varepsilon_a(n) n^{\omega_L(a)} = (-1)^{\omega_L(a)} \omega_L(a)! 2^{L-\omega_L(a) + \sum_{h \in O_L(a)} h}. \quad (3.16)$$

*This is **proved here**. Its sharp-moment core is formalized, and in greater generality: `Fabiús.sum_powerset_neg_one_pow_pow_card (ThueMorseSparseProuhet.lean)` evaluates*

$$\sum_{T \subseteq S} (-1)^{|T|} \left(x + \sum_{j \in T} w_j\right)^{|S|} = (-1)^{|S|} |S|! \prod_{j \in S} w_j$$

for an arbitrary finite index set S , arbitrary step weights w_j in an arbitrary commutative ring, and every offset x . Taking $S = O_L(a)$ and $w_h = 2^h$ produces exactly the constant $(-1)^{\omega_L(a)} \omega_L(a)! 2^{\sum_{h \in O_L(a)} h}$ of (3.16), independently of x , and the companion `Fabiús.sum_powerset_neg_one_pow_eval_of_degree` it is the vanishing (3.15) for every polynomial of degree below $|S|$; `Fabiús.sum_submask_neg_one_pow_pow` is the binary specialization in which S is the bit support of an integer. What is not formalized is the factorization (3.14) itself, nor the free-even-layer step that sums the offset-independent core over the $2^{L-\omega_L(a)}$ even-layer offsets.

Proof. Expand the product by choosing the binary digit $b_h \in \{0, 1\}$ independently. At $z = 1$, an odd exponent contributes a factor $1 - z^{2^h}$ with a simple zero, while an even exponent contributes $1 + z^{2^h} = 2$. Therefore the zero order is exactly ω_L and its leading coefficient in powers of $z - 1$ is

$$(-1)^{\omega_L} 2^{L - \omega_L + \sum_{h \in O_L} h}.$$

The derivatives $T_{a,L}^{(k)}(1)$ are the signed falling-factorial moments. Vanishing for $k < \omega_L$ implies vanishing of ordinary moments of those degrees. At the first surviving degree, lower falling-factorial terms have already vanished, so multiplying the leading coefficient by $\omega_L!$ gives (3.16). $\square$

Remark 3.5. The usual Thue–Morse Prouhet theorem is the case $a_h \equiv 1$, for which $\omega_L = L$. Here the cancellation order is not tied to the block length alone: it is exactly the number of odd spectral layers present before scale L .

4 Bernoulli cumulants, Bell moments, and identifiability

4.1 The same generating function at $q = 4^{-j}$

For $U \sim \text{Unif}[-1/2, 1/2]$,

$$\log \mathbb{E} e^{tU} = \log \frac{\sinh(t/2)}{t/2} = \sum_{j \geq 1} \frac{B_{2j}}{2j} \frac{t^{2j}}{(2j)!}. \quad (4.1)$$

Theorem 4.1 (Cumulant transform). *For an admissible exponent sequence,*

$$\kappa_{2j}(X_a) = \frac{B_{2j}}{2j} A(4^{-j}), \quad \kappa_{2j+1}(X_a) = 0. \quad (4.2)$$

Equivalently,

$$\log \mathbb{E} e^{tX_a} = \sum_{j \geq 1} \frac{B_{2j} A(4^{-j})}{2j} \frac{t^{2j}}{(2j)!}. \quad (4.3)$$

This is [proved here](#).

Proof. Cumulants add under independent sums and scale homogeneously. The level- h contribution to the $2j$ th cumulant is $a_h 2^{-2jh} B_{2j}/(2j)$. Summing over h gives $A(4^{-j})$. $\square$

Corollary 4.2 (Bell-polynomial moments). *The even moments are*

$$\mathbb{E} X_a^{2n} = \mathcal{B}_{2n}(0, \kappa_2, 0, \kappa_4, \dots, 0, \kappa_{2n}), \quad (4.4)$$

or explicitly

$$\mathbb{E} X_a^{2n} = (2n)! \sum_{\substack{j_1, \dots, j_n \geq 0 \\ \sum_{r=1}^n 2j_r = n}} \prod_{r=1}^n \frac{1}{j_r!} \left(\frac{B_{2r} A(4^{-r})}{2r(2r)!} \right)^{j_r}. \quad (4.5)$$

This is a direct, established Bell-polynomial conversion applied to the new master cumulant formula. The conversion itself is formalized: (4.4) is `Fabiús.completeBellPolynomial` (`MomentCumulantAlgebra.lean`), the explicit partition sum (4.5) is `Fabiús.partitionExpSum` (`ExponentialPartition.lean`), and the two are proved to agree by `Fabiús.partitionExpSum_eq_expCoeff` (`ExponentialBell.lean`); all three hold in an arbitrary commutative $\mathbb{Q}$ -algebra, so the formal statement is considerably more general than this application. Its cumulant input [theorem 4.1](#) is not formalized, so the corollary is not yet formally available as stated.

4.2 Zero spectrum and cumulants determine one another

Normalize the even cumulants by

$$c_j(a) = \frac{2j}{B_{2j}} \kappa_{2j}(X_a) = A(4^{-j}). \quad (4.6)$$

Theorem 4.3 (Four equivalent encodings). *The following data determine one another uniquely:*

1. the exponent sequence $(a_h)_{h \geq 0}$;
2. the integer-zero multiplicity function $m_a(n)$;
3. the quotient $\mathcal{Z}_a(s)/\zeta(s) = A(2^{-s})$;
4. the normalized even-cumulant samples $(c_j(a))_{j \geq 1}$.

Constructively,

$$a_v = m_a(2^v) - m_a(2^{v-1}), \quad m_a(2^{-1}) := 0, \quad (4.7)$$

$$a_0 = \lim_{j \rightarrow \infty} c_j, \quad (4.8)$$

$$a_v = \lim_{j \rightarrow \infty} 4^{vj} \left(c_j - \sum_{h=0}^{v-1} a_h 4^{-hj} \right), \quad v \geq 1. \quad (4.9)$$

This is [proved here](#).

Proof. The first formula is the first difference of the prefix sum $m_a(2^v) = \sum_{h \leq v} a_h$. The spectral-zeta equivalence follows from (3.3). For the cumulants, $c_j = A(4^{-j})$, and $4^{-j} \rightarrow 0$ inside the disk of analyticity of A . Equality of all samples for two sequences gives an analytic difference with infinitely many zeros accumulating at the interior point 0, hence the generating functions coincide. The explicit limits follow by peeling off one Taylor coefficient at a time; the remaining tail converges absolutely because $A(1/4) < \infty$. $\square$

Remark 4.4. The limits in (4.8)–(4.9) are mathematically exact but numerically wasteful. The next section replaces them by finite, sign-certified, q -binomial extrapolants.

5 Certified Gaussian- q -binomial Richardson tomography

The coefficient limits in [theorem 4.3](#) are exact, but they use one cumulant at a time and amplify roundoff by powers of 4^j . The geometric nature of the samples admits a much better reconstruction. The result is not merely an acceleration formula: positivity of the hidden coefficients supplies an alternating sign theorem and hence certified integer brackets.

5.1 Lagrange weights on a geometric grid

Fix $q \in (0, 1)$. We use

$$(q; q)_r = \prod_{k=1}^r (1 - q^k), \quad \left[\begin{matrix} r \\ j \end{matrix} \right]_q = \frac{(q; q)_r}{(q; q)_j (q; q)_{r-j}}.$$

For $r \geq 0$ and $0 \leq j \leq r$, let

$$w_{r,j}(q) = \prod_{\substack{0 \leq k \leq r \\ k \neq j}} \frac{-q^k}{q^j - q^k}. \quad (5.1)$$

These are the Lagrange interpolation weights which evaluate, at the origin, a polynomial known at $1, q, \dots, q^r$.

Lemma 5.1 (Closed Gaussian-binomial weights). *One has*

$$w_{r,j}(q) = (-1)^{r-j} \frac{q^{\binom{r-j+1}{2}}}{(q; q)_j (q; q)_{r-j}} = \frac{(-1)^{r-j} q^{\binom{r-j+1}{2}}}{(q; q)_r} \begin{bmatrix} r \\ j \end{bmatrix}_q. \quad (5.2)$$

Consequently,

$$\sum_{j=0}^r w_{r,j}(q) q^{jh} = \delta_{h0} \quad (0 \leq h \leq r). \quad (5.3)$$

For every integer $h \geq r + 1$ the first nonvanishing value is

$$\sum_{j=0}^r w_{r,j}(q) q^{jh} = (-1)^r q^{\binom{r+1}{2}} \begin{bmatrix} h-1 \\ r \end{bmatrix}_q. \quad (5.4)$$

This is **proved here**. The weights (5.1) are formalized as `Fabiús.geometricLagrangeWeight` (`GeometricLagrange.lean`); the annihilation (5.3) is `Fabiús.sum_geometricLagrangeWeight_mul_pow`, with the normalization at $h = 0$ recorded as the genuine boundary value $0^0 = 1$, and the case $h = r + 1$ of (5.4) is `Fabiús.sum_geometricLagrangeWeight_mul_pow_succ`. Both are proved over an arbitrary field, for every q whose powers $q^0, \dots, q^r$ are distinct, so neither uses the hypothesis $q \in (0, 1)$ imposed here. The closed Gaussian form (5.2) and the range $h > r + 1$ of (5.4) are not formalized for general q ; what is available for the latter is `Fabiús.sum_lagrangeEvalWeight_mul_pow_card_add` (`LagrangeResidualMoments.lean`), which expresses every higher moment of an arbitrary distinct-node Lagrange row through a complete homogeneous polynomial, leaving exactly the principal specialization $H_n(1, q, \dots, q^r) = \begin{bmatrix} n+r \\ r \end{bmatrix}_q$ to be supplied.

Proof. Separating the factors with $k < j$ and $k > j$ in (5.1) gives (5.2). The annihilation relations are the exactness of Lagrange interpolation for the monomials $1, x, \dots, x^r$.

For completeness, let $H_h(x_0, \dots, x_r)$ denote the complete homogeneous symmetric polynomial of degree h . The divided difference of x^h at $x_0, \dots, x_r$ is $H_{h-r}(x_0, \dots, x_r)$. The interpolation remainder at $x = 0$, with $x_j = q^j$, therefore gives

$$\sum_{j=0}^r w_{r,j} q^{jh} = (-1)^r q^{0+1+\dots+r} H_{h-r-1}(1, q, \dots, q^r).$$

The principal specialization identity $H_n(1, q, \dots, q^r) = \begin{bmatrix} n+r \\ r \end{bmatrix}_q$ yields (5.4); see, for example, the standard finite q -binomial calculus in [3]. $\square$

5.2 Exact transformed series and sign certification

Let

$$\mathcal{R}_{r,m}[A] := \sum_{j=0}^r w_{r,j}(q) A(q^{m+j}), \quad m \geq 1. \quad (5.5)$$

Scaling all interpolation nodes by q^m leaves the weights unchanged.

Theorem 5.2 (Exact q -Richardson tomography). *Suppose $A(x) = \sum_{h \geq 0} a_h x^h$ converges at $x = q^m$. Then*

$$\mathcal{R}_{r,m}[A] = a_0 + (-1)^r q^{\binom{r+1}{2}} \sum_{h \geq r+1} a_h \begin{bmatrix} h-1 \\ r \end{bmatrix}_q q^{mh}. \quad (5.6)$$

If $a_h \geq 0$, then

$$(-1)^r (\mathcal{R}_{r,m}[A] - a_0) \geq 0. \quad (5.7)$$

Moreover,

$$0 \leq (-1)^r (\mathcal{R}_{r,m}[A] - a_0) \leq \frac{q^{\binom{r+1}{2}}}{(q; q)_r} \left(A(q^m) - \sum_{h=0}^r a_h q^{mh} \right). \quad (5.8)$$

All three statements are **proved here**. The exactness half is formalized: `Fabiús.geometricLagrange_richardson_exact` (`GeometricLagrange.lean`) proves that the weights (5.1) annihilate the first r geometric error modes of a sample block beginning at an arbitrary index and return the constant term exactly. It is stated for samples in an arbitrary module over an arbitrary field, with no positivity hypothesis, no nonzero condition on q and no lower bound on the starting index, so it covers more than the real nonnegative-coefficient setting used here. The exact tail (5.6), the sign (5.7) and the bound (5.8) are not formalized.

Proof. Insert the Taylor series of A into (5.5); absolute convergence permits interchanging the two sums. The terms $1 \leq h \leq r$ vanish by (5.3), while (5.4) gives the remaining terms. The sign follows because Gaussian binomial coefficients are positive for $0 < q < 1$. Finally,

$$\begin{bmatrix} h-1 \\ r \end{bmatrix}_q = \frac{\prod_{k=0}^{r-1} (1 - q^{h-1-k})}{(q; q)_r} \leq \frac{1}{(q; q)_r},$$

which proves (5.8). □

Corollary 5.3 (Certified integer brackets). *If the coefficients are nonnegative integers, then for any odd r_- and even r_+ ,*

$$\lceil \mathcal{R}_{r_-,m}[A] \rceil \leq a_0 \leq \lfloor \mathcal{R}_{r_+,m}[A] \rfloor. \quad (5.9)$$

Once the two rounded endpoints coincide, a_0 has been recovered with a proof certificate. For the Fabiús–Rvachev cumulants, the grid parameter is $q = 1/4$ and the data are $A(q^m) = 2m\kappa_{2m}/B_{2m}$.

5.3 Conditioning does not deteriorate with order

High-order extrapolation is often unstable because its weights become enormous. The geometric grid behaves unusually well.

Theorem 5.4 (Exact ℓ^1 condition number). *The weight norm is*

$$\sum_{j=0}^r |w_{r,j}(q)| = \frac{(-q; q)_r}{(q; q)_r} = \prod_{k=1}^r \frac{1+q^k}{1-q^k}. \quad (5.10)$$

It increases to the finite limit $(-q; q)_\infty / (q; q)_\infty$. At the cumulant grid $q = 1/4$,

$$\frac{(-1/4; 1/4)_\infty}{(1/4; 1/4)_\infty} = 1.96926035366826943194719369697 \dots \quad (5.11)$$

This is **proved here**. Identity (5.10) is formalized only at the half base $q = 1/2$, and is therefore a strict specialization of what is stated here: `Fabiús.sum_abs_discreteLimitWeight` (`FabiúsDiscreteLimitToeplitz.lean`) evaluates the ℓ^1 norm of the r th Toeplitz row as $(-1/2; 1/2)_r / (1/2; 1/2)_r$, and `Fabiús.discreteLimitWeight_eq_geometricLagrangeWeight` identifies that row, after the reversal $j \mapsto r - j$, with the weights (5.1). General q , and hence both the cumulant grid $q = 1/4$ and the limiting value (5.11), is not formalized. The finite q -binomial theorem used in the proof below is however available in full generality as `Fabiús.finite_qBinomial_theorem` (`FiniteqBinomialCore.lean`), over an arbitrary commutative ring and for every q and z , which is exactly the input a general- q proof would need.

Proof. Use (5.2), replace j by $r - j$, and apply the finite q -binomial theorem:

$$\sum_{k=0}^r q^{\binom{k+1}{2}} \begin{bmatrix} r \\ k \end{bmatrix}_q = \prod_{k=1}^r (1 + q^k) = (-q; q)_r.$$

Division by $(q; q)_r$ gives (5.10). Both infinite products converge for $|q| < 1$. $\square$

Remark 5.5. The bounded condition number is a structural advantage over extrapolation on an arithmetic grid. It does not remove the need for accurate high cumulants, but it prevents the interpolation layer itself from introducing an exponentially growing amplification factor.

5.4 Recovering the entire exponent sequence

After $a_0, \dots, a_{v-1}$ have been certified, define the deflated generating function

$$A^{(v)}(x) = x^{-v} \left(A(x) - \sum_{h=0}^{v-1} a_h x^h \right) = \sum_{k \geq 0} a_{v+k} x^k. \quad (5.12)$$

Applying theorem 5.2 to $A^{(v)}$ recovers a_v with the same alternating certificate.

Algorithm 5.6 (Cumulant-to-spectrum tomography). Given certified intervals for the normalized cumulants $c_m = A(4^{-m})$:

1. set $q = 1/4$ and $v = 0$;
2. subtract the already recovered polynomial $\sum_{h < v} a_h q^{mh}$ from each c_m and multiply by q^{-mv} ;
3. evaluate one odd- and one even-order transform (5.5);
4. round outward as in (5.9);
5. increase m or the orders until the integer interval collapses;
6. increment v and repeat.

The truncation estimate (5.8), combined with interval arithmetic, turns the procedure into a fully rigorous recovery algorithm.

Relation to prior Richardson work. The corpus already contains Gaussian- q -binomial acceleration of finite sinc products, dyadic value sequences, spline approximants, and phase-locked endpoint expansions. The new point here is different: the input is the sequence of *even cumulants*, the unknowns are the integer multiplicities of dyadic uniform layers, and nonnegativity turns the Lagrange remainder into alternating spectral certificates.

6 Shift algebra, refinement systems, and rational classification

6.1 The shift is the dilation operator in disguise

Let $(Sa)_h = a_{h+1}$ and $\Delta = S - I$. For signed integer sequences we use the notation Φ_b for the analytic germ at the origin obtained by exponentiating $\sum_h b_h \log \operatorname{sinc}(z/2^h)$; negative entries may make its global continuation meromorphic, but every identity below is unambiguous near zero.

Theorem 6.1 (Master shift–refinement identity). *For every admissible nonnegative sequence,*

$$\boxed{\Phi_a(2z) = \text{sinc}(2z)^{a_0} \Phi_{S_a}(z).} \quad (6.1)$$

More generally, for $m \geq 1$,

$$\Phi_a(2^m z) = \left(\prod_{h=0}^{m-1} \text{sinc}(2^{m-h} z)^{a_h} \right) \Phi_{S^m a}(z). \quad (6.2)$$

Since $S^m = (I + \Delta)^m$, one also has the finite-difference factorization

$$\boxed{\Phi_{S^m a}(z) = \prod_{r=0}^m \Phi_{\Delta^r a}(z)^{\binom{m}{r}}.} \quad (6.3)$$

The final identity is initially an identity of analytic germs; all apparent poles cancel whenever the left side is entire. These statements are **proved here**. The exponent-level content of (6.1)–(6.2)—that passing to $S^m a$ exposes the first m layers and shifts the rest—is formalized as `Fabius.weightedScaleMultiplicity_base_pow_mul` (`WeightedScaleMultiplicity.lean`), for an arbitrary base $b > 1$ and weights in an arbitrary additive commutative monoid. The analytic identities themselves are formalized only at $a_h \equiv 1$, as `Fabius.rvachevFourierProduct_two_mul` and `Fabius.rvachevFourierProduct_two_pow_mul` (`SincProductShells.lean`); these are instances of `Fabius.tprod_geom_scale` and `Fabius.tprod_geom_scale_pow` (`GeometricScaleProducts.lean`), which renormalize a scale product $\prod_n g(z/c^n)$ for an arbitrary factor function g , an arbitrary nonzero ratio c and an arbitrary commutative topological monoid—but for one fixed g , so they do not by themselves reach a variable exponent sequence. The finite-difference factorization (6.3) is not formalized.

Proof. The first formula follows by separating the $h = 0$ factor and reindexing the rest:

$$\prod_{h \geq 0} \text{sinc}(2z/2^h)^{a_h} = \text{sinc}(2z)^{a_0} \prod_{k \geq 0} \text{sinc}(z/2^k)^{a_{k+1}}.$$

Iteration gives (6.2). Finally, exponent sequences add when their transforms multiply, and the binomial identity for the shift gives $S^m a = \sum_r \binom{m}{r} \Delta^r a$. $\square$

Remark 6.2. This formula explains why finite differences, Newton interpolation, and refinement equations continually appear together in the Fabius corpus. The coincidence is not merely an analogy: S acts on the exponent sequence exactly as dilation acts on the sinc product.

6.2 When the refinement tower closes

Write

$$\mathcal{V}(a) = \text{span}_{\mathbb{C}}\{S^j a : j \geq 0\}.$$

The following classical sequence-theoretic equivalence acquires a direct analytic meaning in this setting.

Theorem 6.3 (Rational generating functions are precisely finite refinement systems). *For a complex sequence a of at most exponential growth, the following are equivalent:*

1. $\dim \mathcal{V}(a) < \infty$;
2. a satisfies a constant-coefficient linear recurrence from some point onward;
3. its ordinary generating function $A(q) = \sum_{h \geq 0} a_h q^h$ is rational.

When these conditions hold, the logarithmic transforms $L_j(z) = \log \Phi_{S^j a}(z)$ span a finite-dimensional space stable under $z \mapsto 2z$. Conversely, any finite-dimensional shift-closed exponent model has rational A . This is **proved here** as a synthesis of the standard rational-series theorem with [theorem 6.1](#).

Proof. The equivalence of the first three statements is the usual companion-matrix characterization of rational ordinary generating functions [11]. If $S^d a = \sum_{j < d} \gamma_j S^j a$, then $L_d = \sum_{j < d} \gamma_j L_j$ as analytic germs, while

$$L_j(2z) = a_j \log \operatorname{sinc}(2z) + L_{j+1}(z).$$

Thus the vector $(L_0, \dots, L_{d-1}, \log \operatorname{sinc}(2z))$ obeys a finite refinement system. The converse is the same statement read on the exponent vectors. $\square$

Corollary 6.4 (Pole dictionary for rational weights). *If A is rational, then*

$$\frac{\mathcal{Z}_a(s)}{\zeta(s)} = A(2^{-s}) \quad (6.4)$$

is a finite combination of terms whose poles lie on vertical arithmetic progressions obtained by taking logarithms of the poles of A . Poles of A at roots of unity produce log-periodic terms in Mellin asymptotics; poles away from the unit circle produce additional power-law or exponentially modulated scales. The exact transseries classification suggested by this observation is formulated in [theorem 12.3](#).

7 Polynomial weights and Newton factorization through the Pascal hierarchy

7.1 The audited Pascal–Rvachev transforms

For $r \geq 1$, put

$$\Phi_r(z) := \prod_{h \geq 0} \operatorname{sinc}\left(\frac{z}{2^h}\right)^{\binom{h}{r-1}}. \quad (7.1)$$

The case $r = 1$ is the classical Rvachev transform. The pinned corpus already develops this Pascal hierarchy: compact support and smoothness, refinement ladders, zero divisors, spectral zeta functions, cumulants, Gaussian rank limits, Fourier envelopes, and Mellin endpoint structure. We use those theorems as a basis and ask which other exponent products can be factored through finitely many Φ_r .

7.2 Integer-valued polynomials are exactly the finite Newton sector

Let $P \in \mathbb{Q}[h]$ be integer-valued on $\mathbb{N}_0$ and nonnegative there. Its Newton expansion is

$$P(h) = \sum_{r=0}^d c_r \binom{h}{r}, \quad c_r = \Delta^r P(0) \in \mathbb{Z}. \quad (7.2)$$

The coefficients c_r need not be nonnegative even though every $P(h)$ is.

Theorem 7.1 (Newton–Rvachev factorization). *Let $d = \deg P$ and define $\Phi_P = \Phi_{(P(h))}$. Then*

$$A_P(q) = \sum_{h \geq 0} P(h) q^h = \sum_{r=0}^d c_r \frac{q^r}{(1-q)^{r+1}}, \quad (7.3)$$

$$\boxed{\Phi_P(z) = \prod_{r=0}^d \Phi_{r+1}(z)^{c_r}.} \quad (7.4)$$

Although the right side may be written as a quotient, it is an entire positive-definite function because the unreduced exponent sequence $P(h)$ is nonnegative. Its support radius, integer-zero multiplicity, spectral zeta, and even cumulants are

$$R_P = \frac{1}{2} A_P(1/2) = \sum_{r=0}^d c_r, \quad (7.5)$$

$$m_P(n) = \sum_{r=0}^d c_r \binom{\nu_2(n) + 1}{r + 1}, \quad (7.6)$$

$$\mathcal{Z}_P(s) = \zeta(s) \sum_{r=0}^d c_r \frac{2^s}{(2^s - 1)^{r+1}}, \quad (7.7)$$

$$\kappa_{2j}(X_P) = \frac{B_{2j}}{2j} \sum_{r=0}^d c_r \frac{4^j}{(4^j - 1)^{r+1}}. \quad (7.8)$$

All formulas are **proved here**. Of these, (7.6) is formalized one Newton component at a time: `Fabius.us.weightedScaleMultiplicity_choose` (`WeightedScaleMultiplicity.lean`) proves $m(n) = \binom{\nu_2(n)+1}{r+1}$ for the single Pascal weight $h \mapsto \binom{h}{r}$, for an arbitrary base, its hockey-stick input being `Fabius.inclusivePrefixSum_choose`. Assembling those components into the signed sum over r , and every other formula in the theorem, is not formalized.

Proof. The generating function $\sum_{h \geq 0} \binom{h}{r} q^h = q^r / (1 - q)^{r+1}$ gives (7.3). Exponents add under multiplication, proving (7.4). At $q = 1/2$, every Newton-basis generating function has value 2, which gives (7.5). The hockey-stick identity $\sum_{h=0}^v \binom{h}{r} = \binom{v+1}{r+1}$ proves (7.6); (7.7) and (7.8) follow by substituting $q = 2^{-s}$ and $q = 4^{-j}$ into (7.3). $\square$

Positive-definite quotients. Equation (7.4) gives a systematic source of nonobvious positive-definite quotients of compactly supported entire characteristic functions. Cancellation is certified at the level of dyadic sinc exponents, not guessed from zeros after the fact. This is substantially more flexible than taking ordinary convolutions, which only permits nonnegative exponents in the Pascal basis itself.

7.3 Leading Fourier and endpoint orders

Let $c_d = \Delta^d P(0) > 0$ and set

$$p = d + 2, \quad C_P = \frac{c_d}{(d + 2)! (\log 2)^{d+1}}. \quad (7.9)$$

Lower Newton components have smaller logarithmic order, so the highest Pascal factor controls the leading scale.

Theorem 7.2 (Leading Fourier and small-ball laws). *The sharp Pascal-hierarchy envelope theorem in the audited corpus implies*

$$\limsup_{|x| \rightarrow \infty} \frac{\log |\Phi_P(x)|}{(\log |x|)^{d+2}} = -C_P. \quad (7.10)$$

Let

$$Y_P = X_P + R_P \in [0, 2R_P], \quad G_P(x) = \mathbb{P}(Y_P \leq x).$$

The corresponding exponential Tauberian theorem of de Bruijn type [7, 13] gives

$$-\log G_P(x) \sim C_P (\log(1/x))^{d+2} \quad (x \downarrow 0). \quad (7.11)$$

Consequently the inverse endpoint satisfies

$$\log \frac{1}{G_P^{-1}(y)} \sim \left(\frac{\log(1/y)}{C_P} \right)^{1/(d+2)} \quad (y \downarrow 0). \quad (7.12)$$

These statements are *derived from audited corpus theorems* from the established Pascal hierarchy and the standard exponential Tauberian transfer.

Remark 7.3. For $P(h) \equiv 1$, one has $d = 0$, $c_0 = 1$, and $C_P = 1/(2 \log 2)$, recovering the quadratic logarithmic scale of the classical Fabius endpoint. Increasing the polynomial degree increases the endpoint flatness from a quadratic to a higher logarithmic power.

7.4 The generalized lower-Lambert coordinate

The leading saddle for inverse Laplace analysis solves

$$x = C_P p \lambda^{p-1} e^{-\lambda}. \quad (7.13)$$

It has the explicit lower-branch solution

$$\lambda_P(x) = -(p-1) W_{-1} \left(-\frac{1}{p-1} \left(\frac{x}{C_P p} \right)^{1/(p-1)} \right). \quad (7.14)$$

For $p = 2$ this is the familiar lower-Lambert coordinate of the classical Fabius analysis; the branch notation and inversion conventions follow [6]. For $p > 2$ it gives a single closed coordinate in which the polynomial-weight endpoint transseries should be organized; the Mellin calculation in [section 10](#) explains the periodic coefficients that decorate it. Only the $p = 2$ coordinate is formalized: `Fabius.lowerLambertW` (`LowerLambertW.lean`), a definitional alias of the library’s real lower branch `Real.lambertWLower`, inverts $w \mapsto we^w$ on $(-\infty, -1]$, with functional equation `Fabius.lowerLambertW_mul_exp` and uniqueness `Fabius.lowerLambertW_unique`. The generalized coordinate (7.14) itself has no Lean counterpart.

8 An alternating Newton family of positive-definite Pascal quotients

The abstract Newton factorization becomes more compelling when negative coefficients are unavoidable and the quotient simplifies to something not visibly positive definite.

8.1 Definition and three exact factorizations

For an even integer $d \geq 0$, define the generalized-binomial polynomial sequence

$$P_d(h) = \binom{h-1}{d} \quad (h \in \mathbb{N}_0). \quad (8.1)$$

Here $\binom{-1}{d} = (-1)^d = 1$. Thus

$$P_d(0) = 1, \quad P_d(1) = \cdots = P_d(d) = 0, \quad P_d(h) = \binom{h-1}{d} > 0 \quad (h \geq d+1). \quad (8.2)$$

The parity restriction on d is precisely what makes the $h = 0$ exponent nonnegative.

Theorem 8.1 (Alternating Newton family). *Let $d \geq 0$ be even and put $\Psi_d = \Phi_{P_d}$. Then*

$$A_d(q) = 1 + \left(\frac{q}{1-q} \right)^{d+1}, \quad (8.3)$$

$$P_d(h) = \sum_{r=0}^d (-1)^{d-r} \binom{h}{r}, \quad (8.4)$$

$$\Psi_d(z) = \text{sinc}(z) \Phi_{d+1}(z/2) = \prod_{r=0}^d \Phi_{r+1}(z)^{(-1)^{d-r}}. \quad (8.5)$$

In particular,

$$\Psi_2(z) = \frac{\Phi_1(z) \Phi_3(z)}{\Phi_2(z)}. \quad (8.6)$$

Every expression in (8.5) is the characteristic function of a C_c^∞ probability density supported on $[-1, 1]$. These statements are **proved here**.

Proof. The positive-index part is

$$\sum_{h \geq 1} \binom{h-1}{d} q^h = q \sum_{k \geq 0} \binom{k}{d} q^k = \frac{q^{d+1}}{(1-q)^{d+1}},$$

and the exceptional $h = 0$ term equals 1, proving (8.3). The finite identity (8.4) follows from the backward-shift formula for binomial polynomials, or by comparing coefficients in $(1+t)^{h-1} = (1+t)^h / (1+t)$ through degree d . The first product formula separates $h = 0$ and reindexes $h = k + 1$:

$$\Psi_d(z) = \text{sinc}(z) \prod_{k \geq 0} \text{sinc}\left(\frac{z/2}{2^k}\right)^{\binom{k}{d}} = \text{sinc}(z) \Phi_{d+1}(z/2).$$

The second is [theorem 7.1](#). Since the unreduced exponents (8.2) are nonnegative, Ψ_d is positive definite. Finally, $A_d(1/2) = 2$, so the support radius is one. $\square$

Corollary 8.2 (Concrete random-series model). *Let $U \sim \text{Unif}[-1/2, 1/2]$ and let X_{d+1} have characteristic function Φ_{d+1} , independently of U . Then*

$$Y_d := U + \frac{1}{2} X_{d+1} \quad (8.7)$$

has characteristic function Ψ_d . This exhibits the hidden positivity of the alternating Pascal quotient without any appeal to a zero-by-zero cancellation argument.

8.2 Sparse enhancement of the classical zero divisor

Theorem 8.3 (Exact arithmetic spectrum of Ψ_d). *For every positive integer n ,*

$$\text{ord}_{z=n} \Psi_d(z) = 1 + \binom{\nu_2(n)}{d+1}. \quad (8.8)$$

Its spectral zeta and even cumulants are

$$\mathcal{Z}_d(s) = \zeta(s) \left(1 + \frac{1}{(2^s - 1)^{d+1}} \right), \quad (8.9)$$

$$\kappa_{2j}(Y_d) = \frac{B_{2j}}{2j} \left(1 + \frac{1}{(4^j - 1)^{d+1}} \right), \quad \kappa_{2j+1}(Y_d) = 0. \quad (8.10)$$

In particular,

$$\text{Var}(Y_d) = \frac{1}{12} \left(1 + 3^{-(d+1)} \right). \quad (8.11)$$

These are **proved here**. The hockey-stick step behind (8.8) is formalized as `Fabius.inclusivePrefixSum_choose` (`WeightedScaleMultiplicity.lean`), which proves $\sum_{h \leq v} \binom{h}{d} = \binom{v+1}{d+1}$ for a general Pascal column; together with `Fabius.weightedScaleMultiplicity` it supplies the whole arithmetic content of (8.8) apart from the exceptional $h = 0$ term. The spectral zeta (8.9), the cumulants (8.10) and the variance (8.11) are not formalized.

Proof. Use [theorem 3.1](#) and the hockey-stick identity:

$$\sum_{h=0}^v P_d(h) = 1 + \sum_{h=1}^v \binom{h-1}{d} = 1 + \binom{v}{d+1}.$$

The other formulas are obtained by substituting $q = 2^{-s}$ and $q = 4^{-j}$ into (8.3); the variance is the $j = 1$ case because $B_2/2 = 1/12$. $\square$

Example 8.4 (The first genuinely alternating quotient). For $d = 2$,

$$P_2 = (1, 0, 0, 1, 3, 6, 10, 15, \dots), \quad A_2(q) = 1 + \frac{q^3}{(1-q)^3}.$$

Every odd integer and every integer with $\nu_2(n) \leq 2$ remains a simple zero. The first extra multiplicity occurs at $n = 8$, where the order is 2; at $n = 16$ it is 5; at $n = 32$ it is 11. Yet the same entire function is the quotient $\Phi_1 \Phi_3 / \Phi_2$.

8.3 Automatic lobe signs and sparse Prouhet order

For $h \geq 1$, Lucas' theorem gives

$$P_d(h) \equiv 1 \pmod{2} \iff d \text{ is a binary submask of } h-1. \quad (8.12)$$

This parity sequence is periodic, hence the Fourier-lobe sign sequence of Ψ_d is 2-automatic by [theorem 3.3](#). More precisely, among a complete sufficiently long binary block of values of $h-1$, the forced one-bits of d reduce the density by a factor $2^{-s_2(d)}$.

Corollary 8.5 (Linear Prouhet order with a digital density). *Let*

$$\omega_d(L) = \#\{0 \leq h < L : P_d(h) \text{ is odd}\}.$$

Then

$$\omega_d(L) = 2^{-s_2(d)} L + \mathcal{O}_d(1). \quad (8.13)$$

The length- 2^L lobe-sign block attached to P_d annihilates every power of degree $< \omega_d(L)$, and its first surviving power moment is given exactly by (3.16). Thus the Prouhet order remains linear in L , but its slope is the binary cylinder density selected by d . The Lucas input (8.12) is formalized as `Fabius.odd_choose_iff_bitSupport_subset` (`ThueMorseLucasSupport.lean`), in the submask form “ $\binom{n}{k}$ is odd exactly when the binary support of k is contained in that of n ”, with `Fabius.oddBinomialIndices_eq_image_powerset` parametrizing the odd positions of a Pascal row by the powerset of the bit support. The density count (8.13) is not formalized.

Proof. In each block of 2^k consecutive values of $h-1$, where k exceeds the binary length of d , precisely $2^{k-s_2(d)}$ numbers contain all one-bits of d . Partial end blocks account for the bounded error. The Prouhet statement is [theorem 3.4](#). $\square$

9 Gaussian boundary layers and convergence to a box density

The factorization $Y_d = U + X_{d+1}/2$ makes the large- d limit geometrically transparent. The Pascal variable collapses to zero after normalization, so the new smooth compactly supported density tends to the discontinuous uniform density. The collapse is Gaussian on its natural microscopic scale.

9.1 A quantitative central limit theorem for the Pascal component

Let X_r denote the centered random variable with characteristic function Φ_r . It is the sum of $\binom{h}{r-1}$ independent copies of $2^{-h}U$, with $U \sim \text{Unif}[-1/2, 1/2]$.

Theorem 9.1 (Variance and Berry–Esseen rate). *For $r \geq 1$,*

$$\sigma_r^2 := \text{Var}(X_r) = 3^{-(r+1)}, \quad (9.1)$$

$$\sum \mathbb{E}|\text{summand}|^3 = \frac{1}{4 \cdot 7^r}. \quad (9.2)$$

Hence any valid universal Berry–Esseen constant C_{BE} gives

$$\sup_{y \in \mathbb{R}} \left| \mathbb{P}\left(\frac{X_r}{\sigma_r} \leq y\right) - \Phi_G(y) \right| \leq C_{\text{BE}} \frac{3\sqrt{3}}{4} \left(\frac{3\sqrt{3}}{7}\right)^r. \quad (9.3)$$

This is [proved here](#).

Proof. Since $\text{Var}(U) = 1/12$ and $\mathbb{E}|U|^3 = 1/32$,

$$\begin{aligned} \text{Var}(X_r) &= \frac{1}{12} \sum_{h \geq 0} \binom{h}{r-1} 4^{-h} = \frac{1}{12} \frac{4^{-(r-1)}}{(1 - 1/4)^r} = 3^{-(r+1)}, \\ \sum \mathbb{E}|\text{summand}|^3 &= \frac{1}{32} \sum_{h \geq 0} \binom{h}{r-1} 8^{-h} = \frac{1}{32} \frac{8^{-(r-1)}}{(1 - 1/8)^r} = \frac{1}{4 \cdot 7^r}. \end{aligned}$$

Divide the latter by σ_r^3 and apply the Berry–Esseen theorem for independent, not necessarily identically distributed summands [10]. $\square$

Remark 9.2. The standardized fourth cumulant is also exact:

$$\frac{\kappa_4(X_r)}{\sigma_r^4} = -2 \left(\frac{3}{5}\right)^{r+1}. \quad (9.4)$$

It decays more slowly than the Berry–Esseen upper bound and predicts the first Edgeworth correction in [theorem 12.1](#).

9.2 Boundary scaling of the smooth box approximants

Let f_d be the density of Y_d , and put

$$\tau_d = \frac{1}{2} 3^{-(d+2)/2}. \quad (9.5)$$

This is the standard deviation of $X_{d+1}/2$.

Theorem 9.3 (Gaussian boundary layers). *For each fixed bounded interval of y values,*

$$f_d\left(-\frac{1}{2} + \tau_d y\right) = \Phi_G(y) + \mathcal{O}\left(\left(\frac{3\sqrt{3}}{7}\right)^{d+1}\right), \quad (9.6)$$

$$f_d\left(\frac{1}{2} + \tau_d y\right) = \bar{\Phi}_G(y) + \mathcal{O}\left(\left(\frac{3\sqrt{3}}{7}\right)^{d+1}\right), \quad (9.7)$$

where the implied constant is universal. Away from any shrinking neighborhood of the two endpoints, f_d converges uniformly to the box density $\mathbf{1}_{[-1/2, 1/2]}$. This is [proved here](#).

Proof. Let $T_d = X_{d+1}/2$. Convolution with the unit-height box gives the exact formula

$$f_d(x) = \mathbb{P}(x - 1/2 \leq T_d \leq x + 1/2). \quad (9.8)$$

At $x = -1/2 + \tau_d y$, the lower endpoint of the window lies below the support of T_d for all sufficiently large d , while the upper endpoint is $\tau_d y$. Thus the value is the CDF of T_d/τ_d at y . The right endpoint is the complementary CDF. Apply (9.3). If x stays a distance much larger than τ_d from $\pm 1/2$, the standardized threshold tends uniformly to $\pm\infty$, giving the final assertion. $\square$

9.3 Exact global distances from the limiting box

Theorem 9.4 (Exact L^1 identity and transport bounds). *Let $b(x) = \mathbf{1}_{[-1/2, 1/2]}(x)$. Then*

$$\|f_d - b\|_{L^1(\mathbb{R})} = \mathbb{E}|X_{d+1}|. \quad (9.9)$$

Consequently,

$$\|f_d - b\|_1 \sim \sqrt{\frac{2}{\pi}} 3^{-(d+2)/2}, \quad (9.10)$$

$$d_{\text{TV}}(\text{Law}(Y_d), \text{Law}(U)) \sim \frac{1}{\sqrt{2\pi}} 3^{-(d+2)/2}, \quad (9.11)$$

$$W_2(\text{Law}(Y_d), \text{Law}(U)) \leq \frac{1}{2} 3^{-(d+2)/2}. \quad (9.12)$$

These statements are [proved here](#).

Proof. For a fixed shift $t \in [-1/2, 1/2]$, the two unit intervals $[-1/2, 1/2]$ and $[t - 1/2, t + 1/2]$ overlap in length $1 - |t|$. Thus the shifted box puts mass $|t|$ outside the unshifted box. Averaging over $T_d = X_{d+1}/2$, and using equality of the inside deficit and outside excess, gives

$$\|f_d - b\|_1 = 2\mathbb{E}|T_d| = \mathbb{E}|X_{d+1}|.$$

The normalized variables X_r/σ_r converge to a standard Gaussian by [theorem 9.1](#); their second moments are uniformly one, so the absolute first moments converge as well. This proves (9.10), and total variation is half the L^1 distance. Finally couple $Y_d = U + T_d$ with U itself and use $W_2^2 \leq \mathbb{E}T_d^2 = \tau_d^2$. $\square$

A smooth-to-discontinuous limit with an exact microscopic profile. The family f_d consists entirely of compactly supported C^∞ densities, yet it tends to a discontinuous box. The global error is exactly an absolute moment of a Pascal–Rvachev variable, while the local endpoint transition is a Gaussian survival function. The same Pascal hierarchy therefore controls both the positive-definite quotient and the singular limit it regularizes.

10 Mellin reflection, periodic endpoint hierarchies, and Lambert inversion

10.1 The endpoint Laplace transform

Shift the random series to its left endpoint:

$$Z_a = X_a + R_a = \sum_{h \geq 0} \sum_{j=1}^{a_h} 2^{-h} V_{h,j}, \quad V_{h,j} \sim \text{Unif}[0, 1]. \quad (10.1)$$

Its Laplace transform is

$$L_a(t) = \mathbb{E} e^{-tZ_a} = \prod_{h \geq 0} \left(\frac{1 - e^{-t/2^h}}{t/2^h} \right)^{a_h}. \quad (10.2)$$

Set

$$\lambda(u) = \log \frac{1 - e^{-u}}{u}. \quad (10.3)$$

Then $\log L_a(t) = \sum_h a_h \lambda(t/2^h)$.

Theorem 10.1 (Spectral–endpoint Mellin reflection). *Assume a_h has polynomial growth. On the strip $-1 < \Re w < 0$,*

$$\int_0^\infty \lambda(u) u^{w-1} du = -\Gamma(w) \zeta(w+1), \quad (10.4)$$

$$\int_0^\infty \log L_a(t) t^{w-1} dt = -\Gamma(w) \zeta(w+1) A(2^w). \quad (10.5)$$

Using the spectral zeta from (3.3), this becomes the reflection formula

$$\mathcal{M}[\log L_a](w) = -\Gamma(w) \zeta(w+1) \frac{\mathcal{Z}_a(-w)}{\zeta(-w)}. \quad (10.6)$$

Thus the endpoint asymptotic and the integer-zero divisor are two Mellin views of the same generating function. The contour-transfer interpretation follows the harmonic-sum Mellin framework of [8]. This is **proved here**. Neither (10.4) nor (10.5) is formalized. The library does prove the sibling real-variable integral `Fabius.bose_mellin_integral_zeta` (`MellinBose.lean`), $\int_0^\infty x^{a-1} \log(1 - e^{-x}) dx = -\Gamma(a) \zeta(a+1)$ for real $a > 0$: the same $\Gamma \zeta$ right-hand side, but for the bare kernel $\log(1 - e^{-u})$ and on the half-line to the right of the strip in (10.4), so it is a companion identity and not (10.4) itself. The double pole of $\Gamma(w) \zeta(w+1)$ at $w = 0$ that governs the hierarchy below is regularized by `Fabius.tendsto_gammaZetaRegularProduct_finitePart` (`MellinFinitePart.lean`). The subtraction of $\log u$, the continuation to $-1 < \Re w < 0$, and the complex-variable form all remain to be supplied.

Proof. The kernel behaves as $-u/2 + \mathcal{O}(u^2)$ at zero and as $-\log u + \mathcal{O}(e^{-u})$ at infinity, so the stated strip is maximal for the elementary integral. Integration by parts followed by the classical Mellin integral for $(e^u - 1)^{-1}$ gives (10.4). For (10.5), absolute convergence permits summation under the integral and the substitution $u = t/2^h$ contributes the factor 2^{hw} . Hence the layer sum is $A(2^w)$. The last formula follows from $\mathcal{Z}_a(-w) = \zeta(-w) A(2^w)$. $\square$

10.2 Pole order is the endpoint logarithmic order

Suppose now that $a_h = P(h)$ has degree d and Newton leading coefficient $c_d > 0$. Near $w = 0$, $A_P(2^w)$ has a pole of order $d+1$, while $\Gamma(w) \zeta(w+1)$ has a double pole. At the nonzero dyadic lattice

$$\chi_k = \frac{2\pi i k}{\log 2}, \quad k \in \mathbb{Z} \setminus \{0\}, \quad (10.7)$$

$A_P(2^w)$ has poles of order $d+1$ and the Gamma–zeta factor is regular.

Theorem 10.2 (Polynomial and periodic hierarchy). *As $t \rightarrow \infty$, Mellin inversion of (10.5) gives an expansion of the form*

$$\log L_P(t) = -C_P(\log t)^{d+2} + \sum_{j=0}^{d+1} \alpha_j(\log t)^j + \sum_{k \neq 0} t^{-\chi_k} Q_k(\log t) + \mathcal{E}_P(t), \quad (10.8)$$

where C_P is (7.9), every Q_k is a polynomial of degree at most d , and the remainder comes from the exponentially small endpoint correction and from poles farther to the right after contour displacement. The Fourier coefficients of the periodic-polynomial part are explicit residues of $-\Gamma(w)\zeta(w+1)A_P(2^w)$.

In particular, degree d produces:

- a nonperiodic logarithmic polynomial of degree $d+2$;
- log-periodic coefficient polynomials of degree d ;
- the generalized lower-Lambert saddle (7.14).

This is **proved hereat** the structural level; obtaining a chosen number of explicit coefficients is a finite residue calculation.

Proof. The pole at zero has order $(d+1)+2 = d+3$, and a Mellin pole of order m produces a polynomial of degree $m-1$ in $\log t$. Each nonzero χ_k has order $d+1$ and contributes $t^{-\chi_k}$ times a polynomial of degree d . Since $t^{-\chi_k} = \exp(-2\pi i k \log_2 t)$, these terms are periodic in $\log_2 t$ with polynomial amplitudes. The leading coefficient is inherited from the highest Newton component and agrees with the audited rank- $(d+1)$ Pascal law, giving C_P . $\square$

Remark 10.3. For the classical case $d=0$, the periodic terms are scalar Fourier modes. For $d=1$ they carry affine functions of $\log t$; in general they are degree- d periodic polynomials. This is a genuine hierarchy rather than a cosmetic perturbation of the rank-one Fabius factor.

10.3 Generalized Bernoulli polynomials generate the residues

Define the Nörlund generalized Bernoulli polynomials [9] by

$$\left(\frac{u}{e^u - 1}\right)^\alpha e^{xu} = \sum_{n \geq 0} B_n^{(\alpha)}(x) \frac{u^n}{n!}. \quad (10.9)$$

Every Newton component of $A_P(e^u)$ has the Laurent expansion

$$\boxed{\frac{e^{ru}}{(1 - e^u)^{r+1}} = (-1)^{r+1} u^{-r-1} \sum_{n \geq 0} B_n^{(r+1)}(r) \frac{u^n}{n!}.} \quad (10.10)$$

Algorithm 10.4 (Bernoulli–Bell endpoint coefficient engine). For a polynomial weight P and a desired truncation order:

1. compute its integer Newton coefficients c_r ;
2. expand $A_P(e^u)$ using (10.10);
3. multiply by the Laurent series of $-\Gamma(w)\zeta(w+1)$ with $u = w \log 2$;
4. take the residue at $w = 0$ for the nonperiodic logarithmic polynomial and at $w = \chi_k$ for the periodic coefficient polynomials;
5. solve the saddle recursively in the coordinate (7.14);
6. exponentiate and compose the resulting formal series with complete Bell polynomials.

All ingredients are finite algebra once the desired order and Fourier modes are fixed.

Bernoulli and Bell roles. Ordinary Bernoulli numbers enter the cumulants through (4.2); generalized Bernoulli polynomials encode the higher-order poles of the rational Newton generating function; complete Bell polynomials convert logarithmic cumulants and saddle corrections into moments, amplitudes, and inverse-series coefficients. The three appearances are different faces of one coefficient calculus rather than unrelated special-function decorations.

10.4 Inverse endpoint consequence

Combining (10.8) with saddle inversion yields an all-orders formal expansion for G_P and its inverse in the generalized Lambert coordinate. The leading term was proved in theorem 7.2; the exact periodic-polynomial transseries requires the uniform contour estimates already carried out in the classical corpus to be lifted to higher-order poles. We isolate that analytic completion as theorem 12.4 rather than silently promoting a formal residue calculation to a theorem.

11 Numerical experiments and reproducibility audit

The numerical work was designed to falsify algebraic mistakes, not to replace proofs. The accompanying script `numerical_experiments.py` uses 100-digit arithmetic for the q -Richardson identities and deterministic FFT inversion for the boundary-layer curves. No Monte Carlo sampling is used. The archive contains the script, generated CSV tables, verification log, and both figures.

11.1 Exact identity checks

The script independently verifies:

1. the closed Lagrange weights (5.2);
2. annihilation of $1, q^h, \dots, q^{rh}$ and the exact survivor (5.4);
3. the generating function (8.3) for $d = 0, 2, 4, 6$ at several high-precision arguments;
4. the multiplicity formula (8.8);
5. the complete transformed series (5.6);
6. the condition-number product (5.10).

Every check passed at tolerances between 10^{-65} and 10^{-80} .

11.2 Tomography of the first hidden exponent

For the new $d = 2$ family,

$$A_2(x) = 1 + \left(\frac{x}{1-x} \right)^3, \quad a_0 = 1, \quad q = 1/4.$$

Table 1 records $\mathcal{R}_{r,1}[A_2] - a_0$. The exact alternating sign is visible, and six interpolation steps reduce an initial 3.7×10^{-2} error to 3.6×10^{-16} without a large weight norm.

The limiting interpolation condition number was recomputed as

$$1.96926035366826943194719369697\dots,$$

in agreement with (5.11). This is small enough that the observed convergence reflects the exact tail annihilation rather than a cancellation between huge floating-point weights.

Table 1: Certified q -Richardson errors for A_2 at $m = 1$, $q = 1/4$.

Order r	$\mathcal{R}_{r,1}[A_2] - 1$	Predicted side
0	$3.7037037037 \times 10^{-2}$	upper
1	$-1.1950617284 \times 10^{-2}$	lower
2	$6.9704584427 \times 10^{-4}$	upper
3	$-6.0938671972 \times 10^{-6}$	lower
4	$1.0586844464 \times 10^{-8}$	upper
5	$-4.0454865708 \times 10^{-12}$	lower
6	$3.5620070678 \times 10^{-16}$	upper

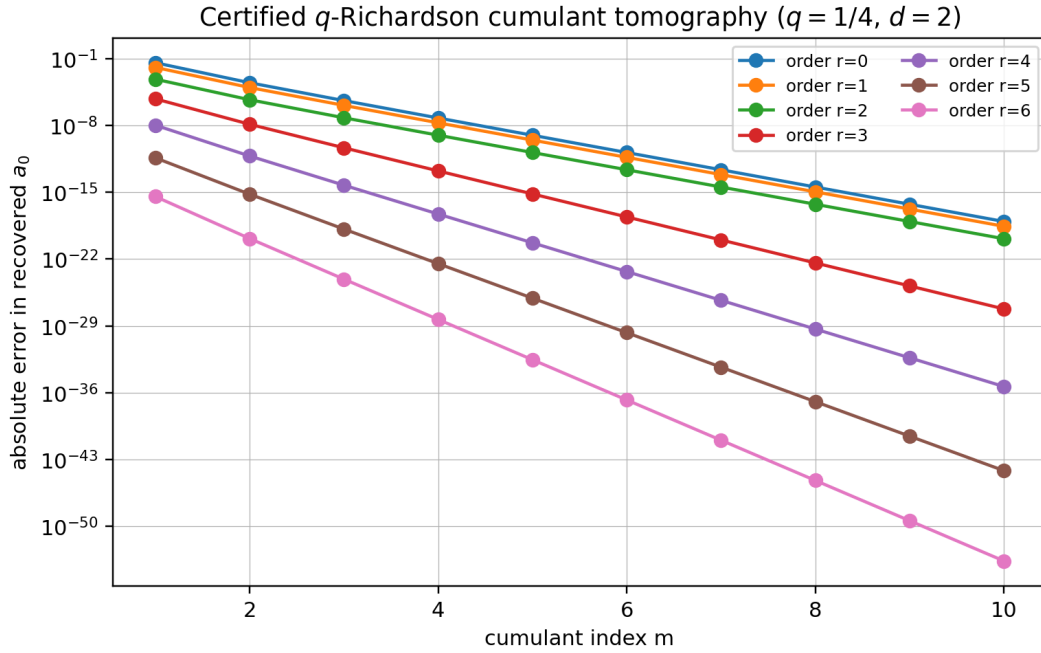


Figure 1: Absolute errors in the recovery of a_0 from normalized cumulants $A_2(4^{-m})$. The CSV file retains the signs; the semilogarithmic plot shows the many orders of magnitude gained by increasing either the cumulant index m or the interpolation order r .

11.3 The sparse zero spectrum for $d = 2$

The generated file `zero_multiplicities_d2.csv` compares the prefix sum of the original exponents with $1 + \binom{\nu_2(n)}{3}$. The first exceptional values are

n	$\nu_2(n)$	zero order
8	3	2
16	4	5
24	3	2
32	5	11
40	3	2
48	4	5
64	6	21

The spectrum is therefore classical at almost every integer but develops rapidly increasing multiplicity on a sparse tower of highly divisible integers.

11.4 Deterministic boundary-layer inversion

For ranks $r = 3, 5, 7, 9$ (equivalently $d = 2, 4, 6, 8$), the script evaluates the exact truncated characteristic product of X_r , inverts it on a nonaliasing FFT interval, normalizes the resulting density, and compares its standardized CDF with the Gaussian. The largest observed CDF errors on $[-4, 4]$ are shown in [table 2](#).

Table 2: Deterministic FFT check of the standardized Pascal CDF.

Rank r	family degree $d = r - 1$	maximum CDF error on $[-4, 4]$
3	2	7.6043×10^{-3}
5	4	2.4624×10^{-3}
7	6	9.3598×10^{-4}
9	8	6.4752×10^{-4}

The errors decrease consistently with [theorem 9.1](#). The theorem’s universal Berry–Esseen bound is intentionally conservative; the exact standardized fourth cumulant (9.4) gives a better explanation of the observed first correction.

11.5 Reproduction command

From the extracted archive, the full experiment can be rerun with

```
python numerical_experiments.py --output-dir reproduced-data
```

The script checks its own exact identities with assertions before writing any figure. A failure therefore stops the run rather than silently producing visually plausible output.

12 Conjectures and frontier problems

The following statements are deliberately separated from the proved theorem package. Each is supported by the exact structure above, but each still requires a substantive analytic, optimization, or arithmetic argument.

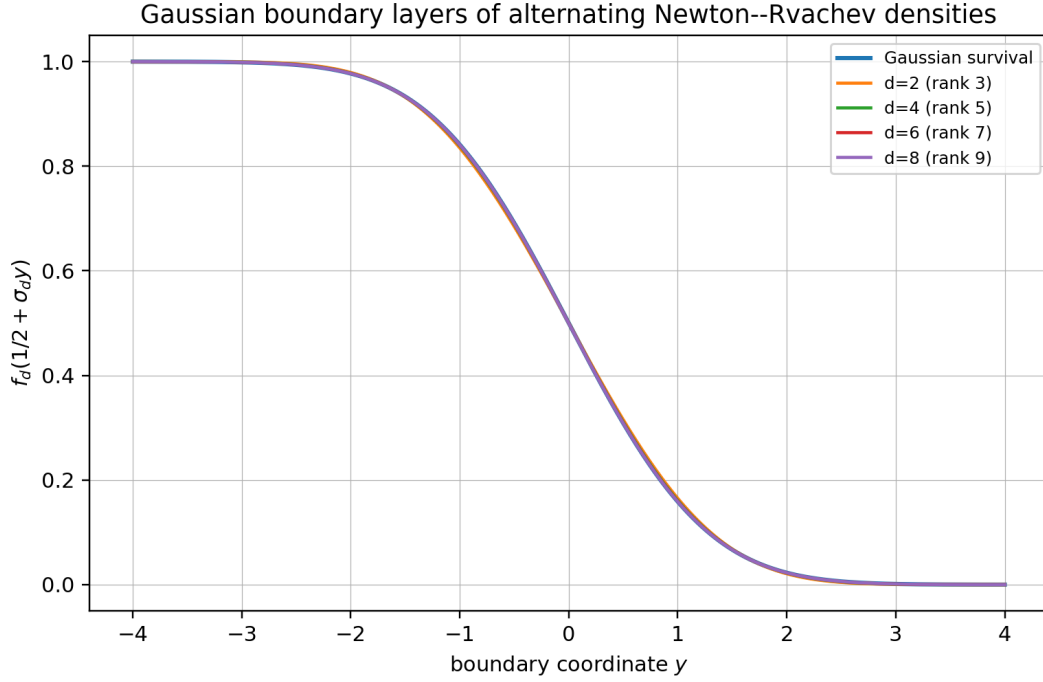


Figure 2: Right-edge boundary profiles $f_d(1/2 + \tau_d y)$ compared with the Gaussian survival function $1 - \Phi_G(y)$. The curves are obtained by deterministic Fourier inversion of the Pascal–Rvachev characteristic products.

Conjecture 12.1 (Uniform Edgeworth boundary expansion). *Let $r = d + 1$ and τ_d be (9.5). For every fixed J and every fixed number of y -derivatives, the two boundary profiles admit a uniform expansion*

$$f_d\left(\frac{1}{2} + \tau_d y\right) = \bar{\Phi}_G(y) + \varphi(y) \sum_{j=1}^J E_j(y; r) + o\left((3/5)^{J(r+1)}\right), \quad (12.1)$$

where φ is the Gaussian density and each E_j is an explicit Hermite polynomial whose coefficients are Bell polynomials in the standardized cumulants. The first correction is governed by $\kappa_4/\sigma^4 = -2(3/5)^{r+1}$.

Why it is plausible. The triangular array has bounded summands, an entire characteristic function, and standardized cumulants that decay exponentially with rank. A Fourier proof should be considerably cleaner than a generic Edgeworth theorem because the exact product provides uniform control in both the central and large-frequency regions.

Conjecture 12.2 (Minimax property of alternating q -tomography). *Fix q, m, r and the cone*

$$\mathcal{C}_M = \left\{ A(x) = \sum_{h \geq 0} a_h x^h : a_h \geq 0, A(q^m) \leq M \right\}.$$

Among all linear estimators using $A(q^m), \dots, A(q^{m+r})$ and exact on polynomials of degree at most r , the Lagrange estimator (5.5) minimizes the worst one-sided error on $\mathcal{C}_M$. Moreover, consecutive odd/even transforms form the narrowest possible universal bracket under these constraints.

Why it is plausible. The remainder kernel in (5.6) has one sign on the coefficient cone and is the unique kernel annihilating the first r monomials. That uniqueness is already formal: Fabiús.eq_1a

`grangeEvalWeight_of_moments` (`GeometricLagrange.lean`) shows that on any finite family of distinct field-valued nodes the Lagrange row is the only row with the prescribed moments, with no nonzeroness or positivity assumption on the competing weights. A proof of the conjecture should be expressible as a finite-dimensional linear-programming duality or a Chebyshev-system extremal argument.

Conjecture 12.3 (Rational refinement transseries). *Let A be rational, nonnegative coefficientwise, and analytic at the origin. Then the complete large- t transseries of $\log L_a(t)$ is determined functorially by the poles of A : each pole ρ of order m produces vertical Mellin lattices*

$$w = \frac{\log \rho + 2\pi i k}{\log 2}, \quad k \in \mathbb{Z},$$

with degree- $(m-1)$ logarithmic amplitudes. Poles on the unit circle yield bounded log-periodic phases; poles of other moduli yield additional power scales. No further frequencies occur.

Research route. Start from (10.5); establish a common meromorphic continuation and vertical bounds for the rational factor, Gamma, and zeta; then shift the Mellin contour across a finite set of pole lattices. The main technical issue is uniform summability of the residue series when several pole lattices have the same real part.

Conjecture 12.4 (All-orders generalized inverse-Fabius expansion). *For every nonnegative integer-valued polynomial P of degree d , the endpoint CDF G_P and its inverse admit all-orders expansions in the generalized lower-Lambert coordinate (7.14). Their coefficients are finite differential polynomials in the degree- d periodic amplitudes of (10.8); equivalently, they are algorithmically generated from Nörlund Bernoulli polynomials, Gamma-zeta jets, and complete Bell polynomials. The expansion is uniform under bounded shifts of the Lambert phase.*

Why it matters. This would extend the corpus's inverse Fabius theory from the rank-one quadratic logarithmic saddle to every polynomial Newton sector and would make the Lambert–Bernoulli–Bell connection exact rather than merely formal.

Open problem 12.5 (The Newton positivity cone). *Classify the integer vectors $(c_0, \dots, c_d)$ for which*

$$P(h) = \sum_{r=0}^d c_r \binom{h}{r} \geq 0 \quad (h \in \mathbb{N}_0).$$

Describe the extreme rays, the indecomposable characteristic functions $\prod_r \Phi_{r+1}^{c_r}$, and the minimal Pascal numerator and denominator after all entire cancellations. Determine whether the alternating vectors $((-1)^d, (-1)^{d-1}, \dots, 1)$ for even d generate distinguished extreme rays under a natural support normalization.

Open problem 12.6 (Arithmetic of generalized half-integer spectra). *For a rational generating function A with integer coefficients, classify the fields, traces, and norms of*

$$\Phi_a \left(k + \frac{1}{2} \right).$$

The audited half-integer report handles the classical exponent sequence. The master zero/cumulant dictionary suggests that rational A should produce finite-dimensional cyclotomic trace recurrences, but the orders and positivity constraints will depend on the pole data of A .

Open problem 12.7 (Optimal smooth approximation of the box within the Newton family). *At fixed support and fixed highest Newton degree, choose a nonnegative integer-valued polynomial P to minimize one of*

$$\|f_P - b\|_1, \quad W_2(\text{Law}(X_P), \text{Law}(U)), \quad \sup_{|\xi| \leq T} |\Phi_P(\xi) - \text{sinc}(\xi)|.$$

The family $P_d(h) = \binom{h-1}{d}$ is natural because it appends one rescaled Pascal law to a box, but it is not yet known to be extremal under any of these criteria.

13 Formalization and computational research program

The theorem package is unusually suitable for staged Lean formalization: its algebraic core is finite, and the analytic results can be added only after the exact identities are stable. A dependency-aware order is as follows.

13.1 Phase I: finite algebra and digital combinatorics

1. Define exponent prefixes and prove $m_a(n) = \sum_{h \leq \nu_2(n)} a_h$ for finite products.
2. Prove the finite zero-count identity $M_a(N) = \sum_h a_h \lfloor N/2^h \rfloor$ and the binary defect (3.8).
3. Define the generalized sign character and prove (3.12); formalize the automaticity equivalence for eventually periodic parity words.
4. Prove the finite sign-product factorization (3.14), its exact order at one, and the first surviving moment (3.16).
5. The geometric Lagrange weights, Gaussian-binomial closed form, exact survivor, and finite residual sign are now formalized; transferring this finite kernel to the report’s analytic Richardson remainder remains open.

These statements require only finite sums, polynomials, valuations, and existing q -binomial infrastructure. Several of them are in fact already discharged in the library, in greater generality than is needed here: step 1 and the floor identity of step 2 by `Fabius.weightedScaleMultiplicity` and `Fabius.sum_range_weightedScaleMultiplicity`; the sharp moment of step 4 by `Fabius.sum_powerset_neg_one_pow_pow_card`, together with its vanishing companion `Fabius.sum_powerset_neg_one_pow_eval_of_degree_lt`; and the annihilation and first surviving moment of step 5 by `Fabius.sum_geometricLagrangeWeight_mul_pow` and `Fabius.sum_geometricLagrangeWeight_mul_pow_succ`, with `Fabius.geometricLagrange_richardson_exact` already giving module-valued Richardson exactness. What genuinely remains in Phase I is the binary defect (3.8), the automaticity equivalence of step 3, the sign-product factorization (3.14) and its free-even-layer count, the closed Gaussian-binomial form (5.2), and the general- q survivor (5.4) with the sign-certified remainder built on it.

13.2 Phase II: probability and entire products

1. Construct X_a from finite uniform sums and pass to the bounded infinite series using the summability hypothesis $A(1/2) < \infty$.
2. Prove local uniform convergence of Φ_a and the convolution-monoid law.
3. Derive cumulants from the uniform Bernoulli expansion and moments from the existing Bell-polynomial library.
4. Formalize identifiability first from zero multiplicities, then from analytic samples $A(4^{-j})$.

5. Build polynomial exponent sequences from Newton expansions and prove the entire cancellation behind (7.4).

13.3 Phase III: asymptotics and inverse functions

1. Reuse the corpus's Mellin kernel theorem to prove (10.5) for polynomial weights.
2. Formalize pole orders and the leading coefficient C_P before attempting the full periodic residue expansion.
3. Transfer the existing Pascal-hierarchy Fourier envelope to a general polynomial by the Newton factorization and lower-order comparison.
4. Prove the leading small-ball and inverse laws. The library contains no de Bruijn–Voss exponential Tauberian transfer, so this step must begin by isolating that theorem as a reusable library component.
5. Only then lift the classical all-orders saddle and inverse-Lambert machinery to the higher-order pole hierarchy.

13.4 Certified computation

The tomography algorithm is a natural application for exact rationals and interval arithmetic. At $q = 1/4$ all Lagrange weights are rational, as are the normalized cumulants of integer exponent sequences. A verified executable can therefore recover coefficients with no floating-point assumptions. For asymptotic experiments, Arb-style ball arithmetic would certify Mellin residues, Lambert branches, and boundary-layer errors. The supplied Python script is a reproducible exploratory implementation, not a substitute for that final certification layer.

14 Conclusion

The central result of this report is a change of viewpoint. Instead of treating the classical Rvachev transform, the Pascal hierarchy, its zero spectrum, its cumulants, and its endpoint asymptotics as separate constructions, one begins with the exponent generating function

$$A(q) = \sum_{h \geq 0} a_h q^h.$$

The same A then appears at four geometrically related arguments:

$$\begin{aligned} A(1/2) &: \text{support radius,} \\ A(2^{-s}) &: \text{spectral zeta and zero divisor,} \\ A(4^{-j}) &: \text{even cumulants,} \\ A(2^w) &: \text{endpoint Mellin transform.} \end{aligned} \tag{14.1}$$

Its coefficient parity is the Fourier-lobe character, its prefix sums are zero multiplicities, its rationality is finite refinement rank, and its Newton coordinates are Pascal–Rvachev exponents.

Three consequences seem especially durable.

First, the spectral and probabilistic data are identifiable: the zero divisor or the full even-cumulant sequence determines the hidden dyadic convolution law. The Gaussian- q -binomial Richardson transform makes that statement constructive and supplies certified alternating bounds.

Second, nonnegative integer-valued polynomials form a rich positive-definite quotient sector. Negative Pascal exponents are not pathologies when they recombine into a nonnegative scale sequence. The family $P_d(h) = \binom{h-1}{d}$ gives the explicit quotient $\Phi_1 \Phi_3 / \Phi_2$ at $d = 2$, sparse enhanced zero divisors, automatic lobe signs, and a smooth approximation to the box with an exact Gaussian edge profile.

Third, Mellin reflection makes the endpoint and spectrum literally the same meromorphic object viewed at s and $-s$. Polynomial degree becomes pole order; pole order becomes the power of the endpoint logarithm; the dyadic pole lattice becomes the periodic coefficient hierarchy; and the leading saddle is solved by a generalized lower branch of Lambert W . Nörlund Bernoulli polynomials and complete Bell polynomials then supply the coefficient algebra required for an all-orders inverse theory.

These deductions are new relative to the pinned 67-source corpus, but they are not presented as the end of the subject. Their main value is that they turn a collection of striking special identities into a classification problem with exact algebraic coordinates, stable reconstruction algorithms, and a clear route to formal verification.

A Complete recursive corpus inventory

The following path ledger is the authoritative source list used for the nonduplication audit. Every path is relative to `Analysis/FabiusFunction/docs/`. The pinned commit is `32d6d36c51d803289e6d6a0dc0c37753766eba47`. The count is 67. Frozen revisions were read for distinctive claims and provenance, but a repeated theorem in several revisions was counted once when determining novelty.

No.	Relative path
1	FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
2	Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
3	Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
4	fabius_lean_walkthrough/fabius_lean_walkthrough.tex
5	non-formalized-research-frontiers/non-formalized-research-frontiers.tex
6	archive/early-notes/FabiusAsymptotic/FabiusAsymptotic.tex
7	archive/early-notes/K-foldsummationoverthesignedThue-Morsesequence/K-foldsummationoverthesignedThue-Morsesequence.tex
8	archive/glossary/FabiusFunction_Mathematical_Glossary-2/FabiusFunction_Mathematical_Glossary.tex
9	archive/glossary/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
10	archive/lean-walkthrough/Fabius_Function_Lean_Walkthrough/Fabius_Function_Lean_Walkthrough.tex
11	archive/lean-walkthrough/fabius_lean_walkthrough/fabius_lean_walkthrough.tex
12	archive/primary-exposition/Fabius_Function_and_Rvachev_Up-2/Fabius_Function_and_Rvachev_Up.tex
13	archive/primary-exposition/Fabius_Function_and_Rvachev_Up-3/Fabius_Function_and_Rvachev_Up.tex
14	archive/primary-exposition/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
15	archive/primary-exposition/Fabius_and_Rvachev_Up-2/Fabius_and_Rvachev_Up.tex
16	archive/primary-exposition/Fabius_and_Rvachev_up/Fabius_and_Rvachev_up.tex
17	archive/primary-exposition/Fabius_function_and_up_function/Fabius_function_and_up_function.tex
18	archive/primary-supplements/Fabius_Function_Missing_Parts-2/Fabius_Function_Missing_Parts.tex
19	archive/primary-supplements/Fabius_Function_Missing_Parts/Fabius_Function_Missing_Parts.tex

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Corpus inventory (continued)

No.	Relative path
20	archive/q-calculus/Fabius_q_Special_Functions/Fabius_q_Special_Functions.tex
21	archive/q-calculus/fabius-q-special-functions/fabius-q-special-functions.tex
22	archive/research-frontiers/Fabius_Dyadic_Asymptotic_Bridge/Fabius_Dyadic_Asymptotic_Bridge.tex
23	archive/research-frontiers/Fabius_Dyadic_Formulae_and_Alternative_Representations/Fabius_Dyadic_Formulae_and_Alternative_Representations.tex
24	archive/research-frontiers/Fabius_Dyadic_Formulae_to_Asymptotics/Fabius_Dyadic_Formulae_to_Asymptotics.tex
25	archive/research-frontiers/Fabius_Dyadic_q_Connections/Fabius_Dyadic_q_Connections.tex
26	archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rate/Fabius_Thue_Morse_Convergence_Rate.tex
27	archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rates/Fabius_Thue_Morse_Convergence_Rates.tex
28	archive/research-frontiers/Primary_Exposition_Gap_Register/Primary_Exposition_Gap_Register.tex
29	archive/research-frontiers/Repeated_Integration_and_Rvachev_Up/Repeated_Integration_and_Rvachev_Up.tex
30	archive/research-frontiers/Rvachev_Up_from_Repeated_Integration/Rvachev_Up_from_Repeated_Integration.tex
31	archive/research-frontiers/Thue_Morse_Formula_Atlas-1/Thue_Morse_Formula_Atlas.tex
32	archive/research-frontiers/Thue_Morse_Formula_Atlas-2/Consolidated_Formula_Atlas_for_the_Thue_Morse_Sequence.tex
33	archive/research-frontiers/Thue_Morse_Formula_Atlas-3/Thue_Morse_Formula_Atlas.tex
34	archive/research-frontiers/non-formalized-research-frontiers/non-formalized-research-frontiers.tex
35	archive/standalone-studies/Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
36	archive/standalone-studies/Small_Argument_Asymptotics/Small_Argument_Asymptotics.tex
37	archive/standalone-studies/fabius-inverse-dyadic-closed-form/fabius-inverse-dyadic-closed-form.tex
38	non-formalized-research-frontiers/drafts/Fabius_Arithmetic_Rays_Frontier_Report/Fabius_Arithmetic_Rays_Frontier_Report.tex
39	non-formalized-research-frontiers/drafts/Fabius_Half_Integer_Spectral_Frontier_Report/Fabius_Half_Integer_Spectral_Frontier_Report.tex
40	non-formalized-research-frontiers/drafts/Fabius_Rvachev_Frontier_Report-2/Fabius_Rvachev_Frontier_Report.tex
41	non-formalized-research-frontiers/drafts/Fabius_Rvachev_Frontier_Report/Fabius_Rvachev_Frontier_Report.tex
42	non-formalized-research-frontiers/drafts/Fabius_Rvachev_New_Frontiers/Fabius_Rvachev_New_Frontiers.tex
43	non-formalized-research-frontiers/drafts/Fabius_Rvachev_Thue_Morse_Frontier_Results/Fabius_Rvachev_Thue_Morse_Frontier_Results.tex
44	non-formalized-research-frontiers/drafts/Spectral_Arithmetic_Pascal_Rvachev_Hierarchy/Spectral_Arithmetic_Pascal_Rvachev_Hierarchy.tex
45	non-formalized-research-frontiers/drafts/Thue_Morse_Formula_Atlas/Thue_Morse_Formula_Atlas.tex
46	non-formalized-research-frontiers/drafts/fabius_frontier_new_results/fabius_frontier_new_results.tex

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Corpus inventory (continued)

No.	Relative path
47	non-formalized-research-frontiers/drafts/fabius_frontier_results_bundle/fabius_frontier_results.tex
48	non-formalized-research-frontiers/drafts/fabius_frontier_spectral_endpoint_report_bundle/fabius_frontier_spectral_endpoint_report.tex
49	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-2/Rvachev_Up_Fourier_Decay.tex
50	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-3/Fourier_decay_of_Rvachev_up.tex
51	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Comparative_Audit/Rvachev_Up_Fourier_Decay_Comparative_Audit.tex
52	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Gentle_Guide/Rvachev_Up_Fourier_Decay_Gentle_Guide.tex
53	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Second_Wave_Audit/Rvachev_Up_Fourier_Decay_Second_Wave_Audit.tex
54	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions.tex
55	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-1/rvachev_up_fourier_decay.tex
56	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-4/rvachev_up_fourier_decay.tex
57	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-5/Rvachev_Up_Fourier_Decay_Final.tex
58	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-6/Rvachev_Up_Fourier_Decay_Final_Synthesis.tex
59	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-7/rvachev_up_fourier_decay_final.tex
60	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-8/rvachev_up_fourier_decay_final.tex
61	papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0-corrected.tex
62	papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0.tex
63	papers/arXiv-1502.06738v2/Lacunary_products.tex
64	papers/arXiv-1702.05442v1/09-Function.tex
65	papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex
66	papers/arXiv-2211.13570v1/document.tex
67	papers/ubiq15/ubiq15-corrected.tex

B Result ledger and dependency map

Table 4 records the logical status and principal inputs of the report’s main claims. It is intended to make later formalization and peer checking easier: a reader can separate finite algebra from analytic continuation, and new deductions from imported frontier theorems.

Table 4: Result and proof-dependency ledger.

ID	Statement	Status	Main inputs
A1	Exponent-sequence probability law, support, entire product, convolution monoid	proved	uniform characteristic function; normal convergence
A2	Zero multiplicity $m_a(n)$ and zeta $\zeta(s)A(2^{-s})$	proved	Euler product; Tonelli/rearrangement
A3	Digital zero defect and generalized lobe character	proved	binary floors; parity
A4	Automaticity iff the parity word is eventually periodic	proved	exact 2-kernel/shift identity
A5	General Prouhet order and first surviving power moment	proved	finite product; falling factorials
B1	Cumulant transform $B_{2j}A(4^{-j})/(2j)$	proved	uniform cumulants; additivity
B2	Identifiability from zeros, zeta, or all even cumulants	proved	prefix differences; identity theorem
B3	Gaussian- q -binomial tomography and one-sided remainder	proved	Lagrange interpolation; principal specialization
B4	Bounded exact condition number $(-q; q)_r/(q; q)_r$	proved	finite q -binomial theorem
C1	Shift/refinement identity and rational finite-rank classification	proved	reindexing; rational-series theorem
C2	Newton factorization of nonnegative polynomial weights	proved	integer-valued polynomial calculus
C3	Leading Fourier, small-ball, and inverse laws	derived	audited Pascal hierarchy; de Bruijn transfer
D1	Alternating family $P_d(h) = \binom{h-1}{d}$ and positive quotient	proved	Newton identity; random-series factorization
D2	Sparse spectrum, exact cumulants, automatic Prouhet selector	proved	hockey-stick; Lucas theorem
D3	Gaussian edge layer, exact L^1 law, TV and W_2 bounds	proved	Berry–Esseen; interval overlap
E1	Mellin reflection between endpoint Laplace and spectral zeta	proved	Mellin kernel; geometric scaling
E2	Polynomial/log-periodic pole hierarchy	proved	Mellin residues; rational pole orders
E3	All-orders generalized inverse transseries	structurally conjectural	uniform higher-pole saddle analysis needed

Lean crosswalk for B3–B4. The finite weight, all-moment, residual-sign, and finite-norm identities in B3–B4 are formalized in `GeometricQBinomialLagrange.lean`, `GeometricLagrangeQBinomial.lean`, and `GeometricLagrangeQMoments.lean`. The tomography interpretation, one-sided analytic remainder, and infinite condition-number limit remain results of this report rather than claims about the current Lean corpus.

C Reproducibility manifest

The companion archive contains:

Path	Purpose
Exponent_Sequence_Newton_Rvachev_Frontiers.tex	Complete manuscript source.
Exponent_Sequence_Newton_Rvachev_Frontiers.pdf	Rendered report.
numerical_experiments.py	Commented high-precision and FFT verification program.
figures/richardson_tomography.png	Figure used in figure 1 .
figures/boundary_layers.png	Figure used in figure 2 .
data/richardson_tomography.csv	Full signed extrapolation errors.
data/boundary_layer_errors.csv	Deterministic standardized-CDF errors.
data/zero_multiplicities_d2.csv	Direct and closed-form zero orders for the $d = 2$ family.
data/verification_log.txt	Precision settings, assertion status, and headline values.
README.md	Build and reproduction commands.
requirements.txt	Exact Python package versions used for the numerical audit.
SHA256SUMS	Checksums for the report, code, figures, and data.

The numerical script fixes the precision and grid sizes explicitly. Its formulas use `numpy.sinc(x) = sin( $\pi x$ ) / ( $\pi x$ )`, matching the normalization in (2.1). The FFT interval is wider than the compact support, so periodic copies do not overlap. All exact checks run before figure generation.

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